

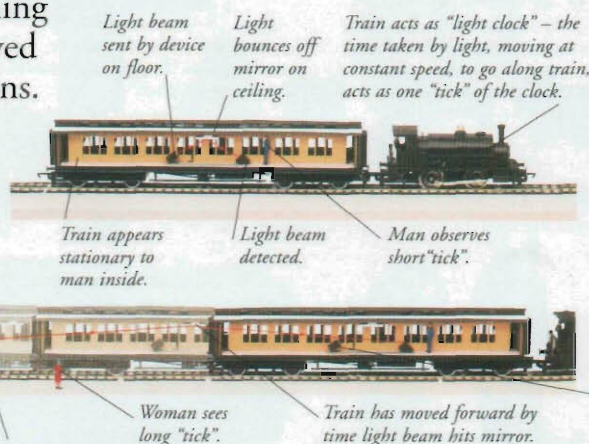
EINSTEIN, ALBERT

$$E=mc^2$$

ALBERT EINSTEIN WAS a scientific genius who changed the way we view our universe. In 1905, he united space and time in one mathematical description. Ten years later he proposed a complete theory of gravity that explained how the universe works, relating mass and energy in the famous equation $E = mc^2$. Many people doubted his theories, but later investigation has since proved Einstein's theories to have been correct. As well as transforming the science of physics, Einstein's work paved the way for the creation of nuclear weapons.

Special Theory of Relativity

In the early 1900s, Einstein developed the Special Theory of Relativity. This says that time is relative: it passes differently for individuals, depending on how fast or slowly they move. The faster anything travels, the slower time seems to pass. If one person travels into space close to the speed of light and another stays on Earth, time passes slower for the person in space. On their return, the person on Earth will be older.



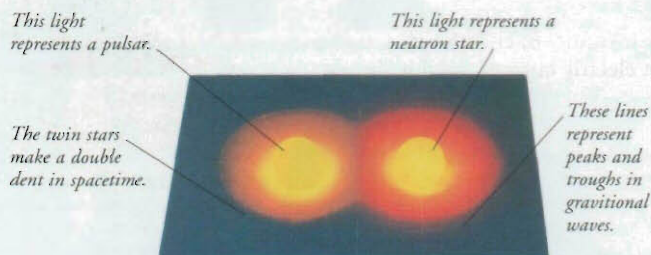
Early life

Einstein was born in Ulm, Germany, and studied in Switzerland before graduating from Zurich's Institute of Technology in 1900. He did not fit in at school because he asked many difficult questions, and could get no work until he found a job in the Patent Office in Bern in 1902.



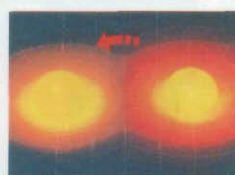
Moving clocks

According to the special theory, time measured by a moving clock will run slower than if measured by a stationary clock. This can be demonstrated by light beams carried on a train travelling at nearly the speed of light. A person on the train sees the light travel a short distance; an observer on the platform sees it travel further because of the train's movement.

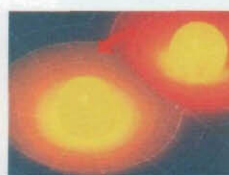


Ripples in space

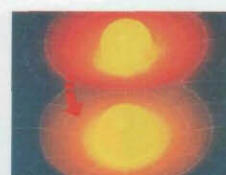
Einstein's theory predicted that objects jiggling around in space – such as two stars in a binary pulsar system – would make ripples in space. These ripples can be detected as gravitational waves. Subsequent experiments have proved Einstein's theory correct.



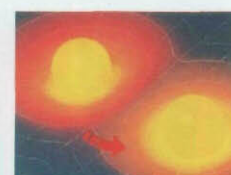
Stars rotate anti-clockwise.



Neutron star moves around pulsar.



Stars' positions change in relation to observer.



Stars continuously swap places.

General Theory of Relativity

Einstein developed the General Theory of Relativity that explained gravity and the nature of space. He explained that as light travels the shortest path through space, when it bends space must be curved. Planets that travel round the Sun are thus following a straight a path as possible through curved space.

Making waves

Stars in a binary pulsar rotate round each other. As they move, they make waves in space. The waves carry energy from the stars, causing the stars to slow down as they lose energy. The rate that a pulsar slows in its orbit exactly matches Einstein's theory, though the first pulsar was not discovered until 1968.

Mileva Einstein

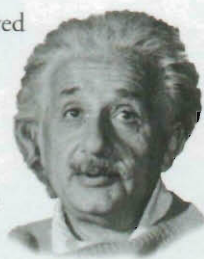
Einstein married his first wife Mileva, a mathematician and scientist in 1903. They had a daughter and two sons. Mileva worked closely with her husband and helped with his research, though to what degree she influenced his work is unknown. They were divorced in 1919.



Mileva and her son Hans Albert

Political life

In 1933, Einstein moved to America to avoid Nazi persecution as a Jew, and campaigned for a Jewish state. He realized that his theories made possible the creation of nuclear weapons, but campaigned against such weapons after World War II. In 1952, he was offered the presidency of Israel, but declined the offer.



The bomb

In the late 1930s, Einstein feared that Nazi Germany would use nuclear weapons in war, so he wrote to US president Franklin D. Roosevelt in 1939, urging the USA to begin constructing atomic weapons to counter this threat.



Explosion of atomic bomb

ALBERT EINSTEIN

- 1879 Born in Ulm, Germany.
- 1896–1900 Studies at Institute of Technology, Zurich, Switzerland.
- 1902–9 Works in Patent Office, Bern, Switzerland.
- 1905 Obtains doctorate; writes Special Theory of Relativity.
- 1914 Moves to Berlin.
- 1915 Writes General Theory of Relativity.
- 1921 Awarded Nobel Prize for Physics.
- 1933 Moves to the USA.
- 1952 Offered presidency of Israel.
- 1955 Dies in Princeton, USA.

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